

Advancing a Framework for Grid Reliability Incentives in Low- and Middle-Income African Countries

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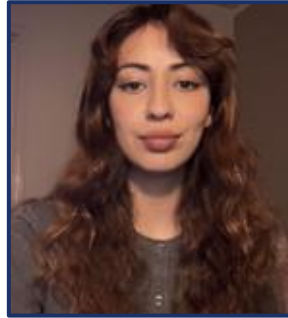
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Introduction

Focus on Reliable Connections

- Over **600 million** Africans still lack reliable electricity access
- **Mission 300**: A World Bank and African Development Bank initiative targeting to provide electricity to **300 million** Africans by 2030
- Despite growing connections, **reliability remains a bottleneck** due to:
 - ❖ Inadequate and harder measurement of performance
 - ❖ Limited enforcement of service standards
 - ❖ Institutional and regulatory constraints



Objective



Incentivize reliability improvements through market-based mechanisms in LMIC



Assess reliability in Africa



Identify pilot countries



Develop an intervention
proposal

Research Questions

Primary questions:

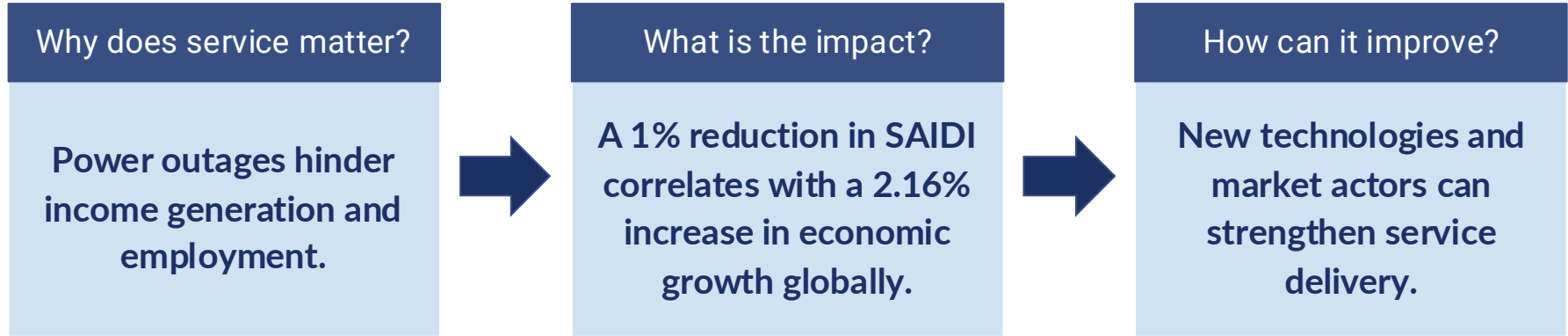
- Where in sub-Saharan Africa should a performance-based reliability pilot be launched?
- How should it be structured to last beyond the project?

Supporting question:

- What do existing incentive programs tell us about what works, what fails, and why?



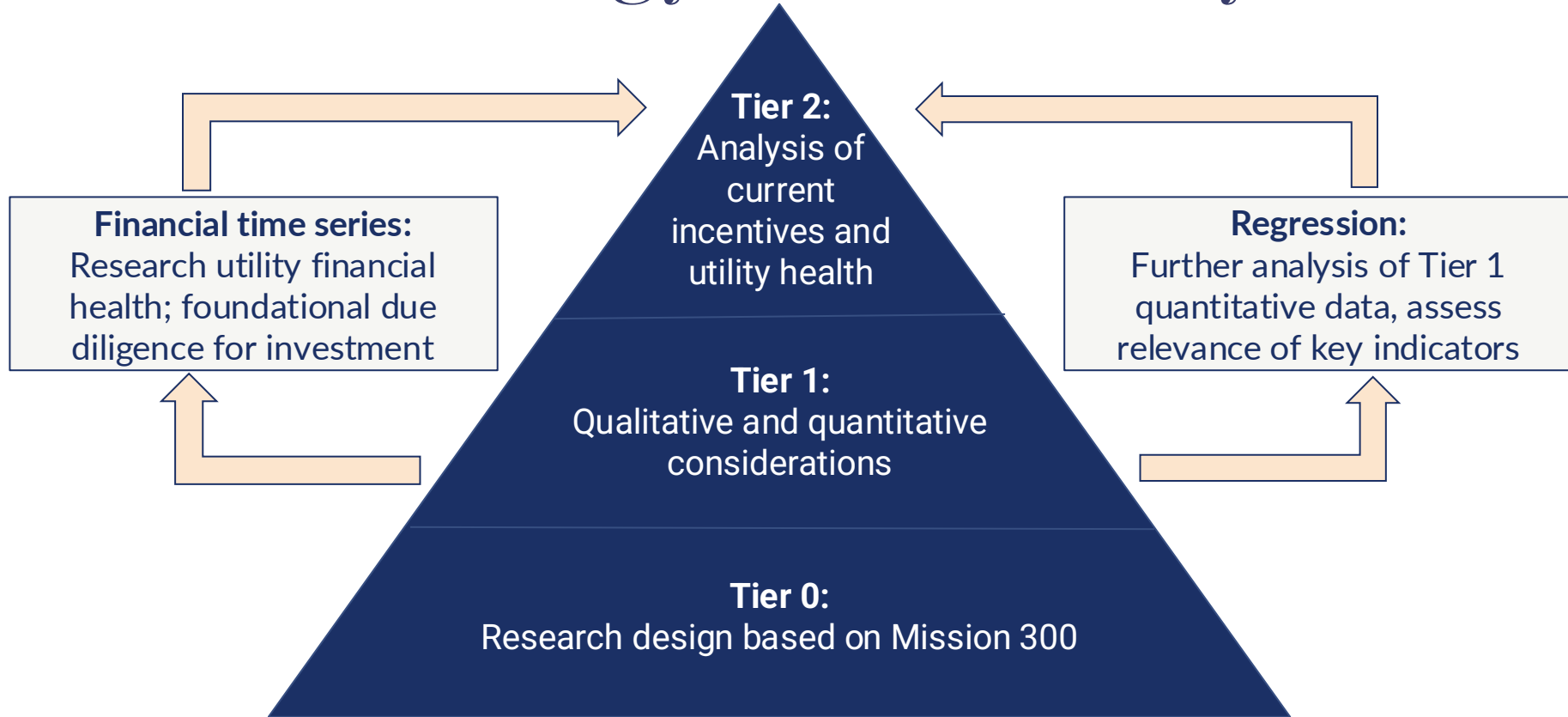
Literature Review



Client Proposal: Performance-Linked Reliability Program:

- Diagnose causes of unreliability using data
- Implement targeted utility improvements
- Reward verified reliability improvements through performance payments

Methodology: Tiered Analysis



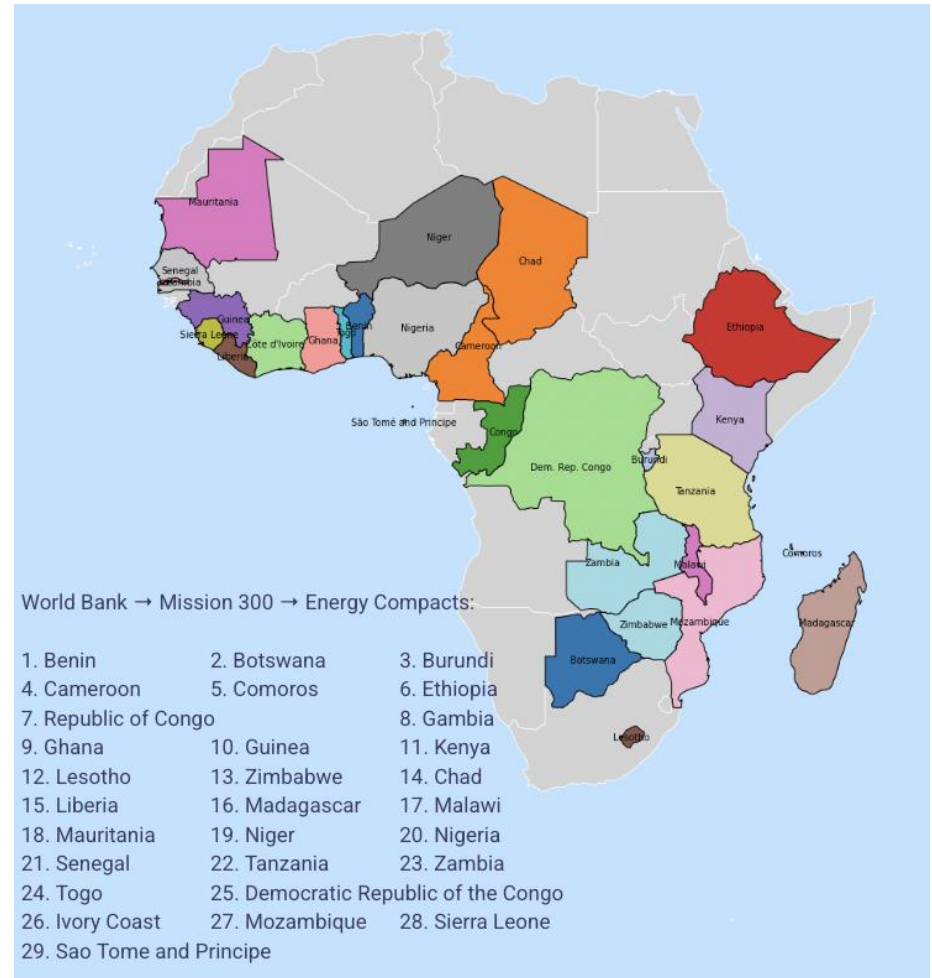
Tier 0

Africa has 54 fully recognized states

→ How can we select the optimal pilot country?

Mission 300 as Screening Criterion:

- Participate in Mission 300
- Published a National Energy Compact



Tier 1 Methodology

Quantitative Indicators / Generation Adequacy Indicators

**Installed and Available
Capacity vs Peak Demand**

Reliability metrics:

- 1. SAIFI**
- 2. SAIDI**
- 3. Load shedding hours**

Technical Losses

Qualitative Criteria

**Political
Commitment to
Reliability**

**Regulatory
Autonomy**

**Public
Demand/Opinion**

**Data Availability
and Research
Constraints**

Analyses: Regression and Time Series

Regression

- **Identify relationships between reliability metrics and other indicators**
 - SAIDI, SAIFI v Technical losses (%)
 - SAIDI, SAIFI v Normalized Capacity Gap
- **Normalized Capacity Gap**
 - $(\text{Peak Demand} - \text{Available Capacity}) / \text{Peak Demand}$

Time Series

- SAIDI, SAIFI from 2012-2023
- Dataset: Utility Performance and Behavior in Africa Today (UPBEAT)

Tier 1: Results

Quantitative

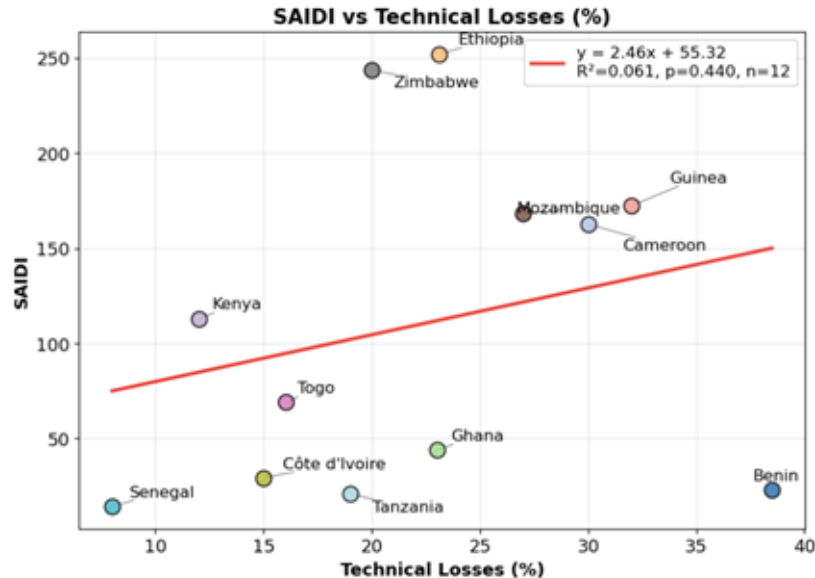
Qualitative

Nation	Installed Capacity (MW)	Available Capacity (MW)	Peak Demand (MW)	Technical Losses (%)	Load Shedding (hrs/mth)	SAIFI	SAIDI	Political Commitment	Regulatory Autonomy	Public Demand	Data availability
Ethiopia	5656.5	4242.37	4916	23.1	21	468	252	Moderate	Moderate	High	High
Ghana	5,260	4,855	4,125	23	0	27	44	High	Moderate	High	High
Kenya	3840.8	2,316	2439	12	150	44	113	High	Moderate	High	High
Togo	254	200	304.6	16.04	N/A	33	69	High	Moderate	N/A	High
Côte d'Ivoire	2,548	N/A	1807.7	15	N/A	22	29.37	High	Moderate	N/A	High
Nigeria	13000	5,395.70	20000	37.92	67.2	N/A	N/A	High	Moderate	High	High
Tanzania	3404	2011	1500	19	N/A	46.8	20.9	Moderate	Low	High	High

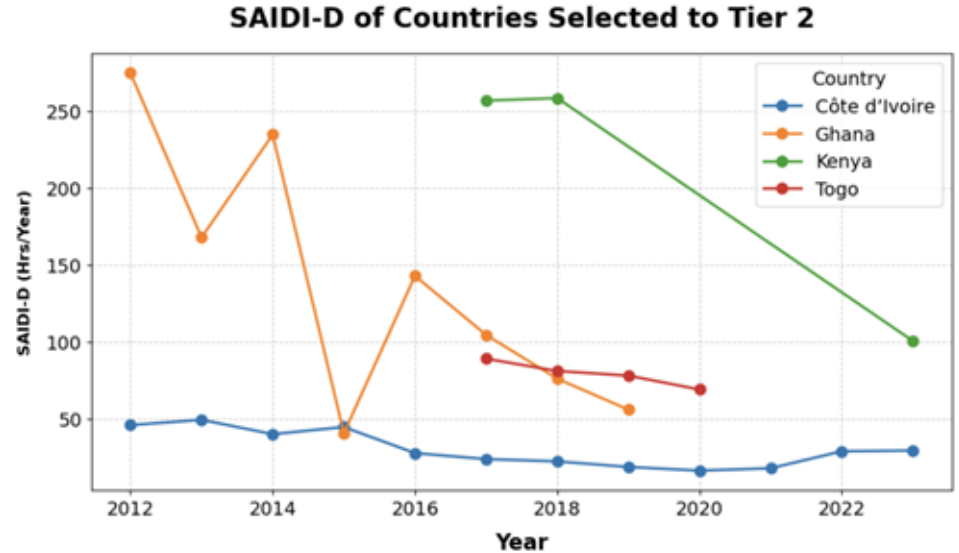
Results: Regression Model and Time Series

Regression: SAIDI by Technical Losses %

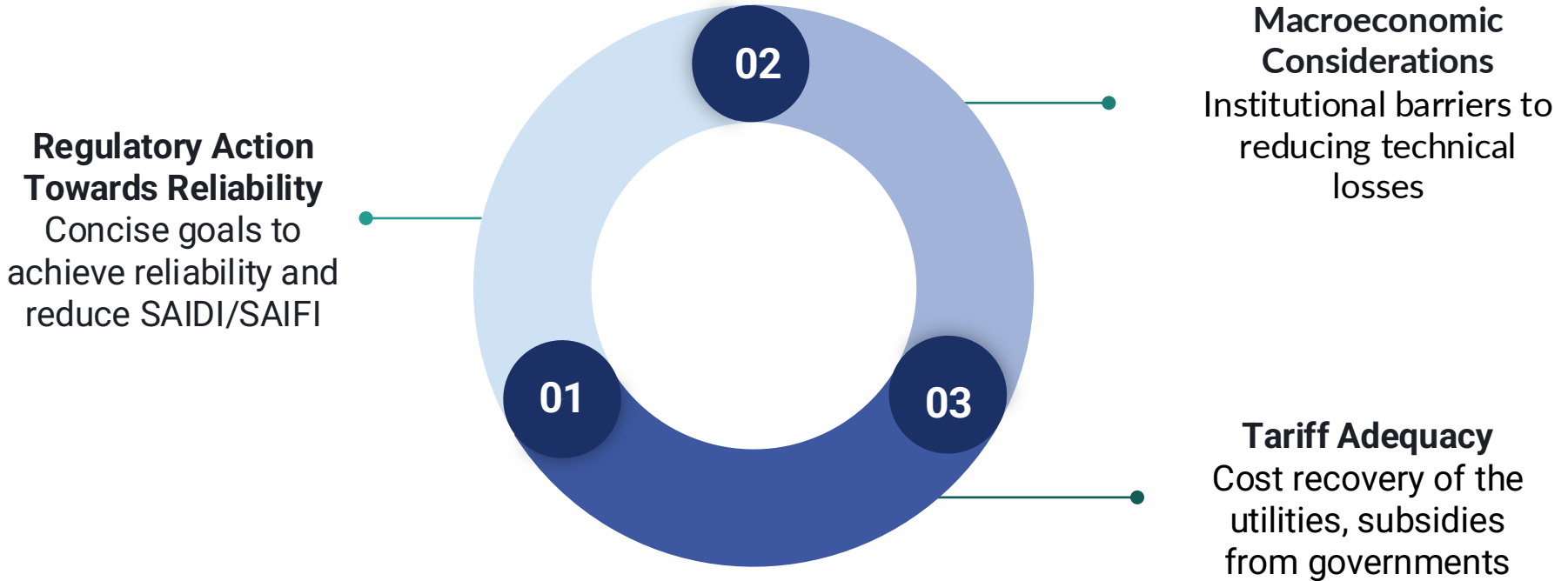
Results are insignificant: $R^2 = 0.061$



Time Series: SAIDI-D from 2012-2023



Tier 2 Analysis



Tier 2: Key Insights

Financial health is a precondition for reliability — not an outcome

Tariff reform is the single most common structural barrier

The reliability problem is a distribution problem, not a generation problem

Performance gains must be maintained, not just achieved

No one-size-fits-all: differentiated intervention structures are needed

Results: Utility Finances & Economics

Sub-Saharan Africa

Low
Collection Rate
+ subsidized tariffs

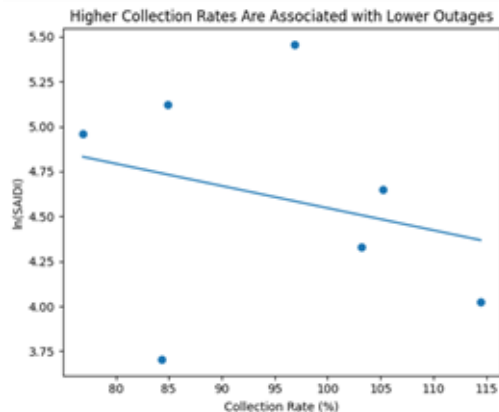
< 1.0x
Median DSCR
(most years)

Reversed
SAIDI gains post-
2021 fragile

Institutional conditions

- Utilities are overleveraged
- Cost recovery low
- Reliant on government subsidies
- Low or no private sector investment

Higher collection rate ~ lower outages



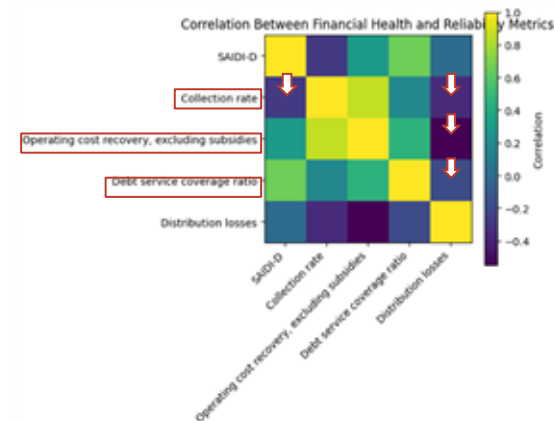
Insights

Policy reform alone insufficient.

Structured KPI-linked investing system to:

1. Remove institutional barriers

2. Strengthen what's working



Proposed Cap Stack

KPI TRIGGERS:

Collection Rate

Loss Reduction

Transmission Availability

SAIDI/SAIFI Reduction

UNLOCKED CAPITAL

Grant & First-Loss Capital	10-25%	MIGA / Sov. Guarantee
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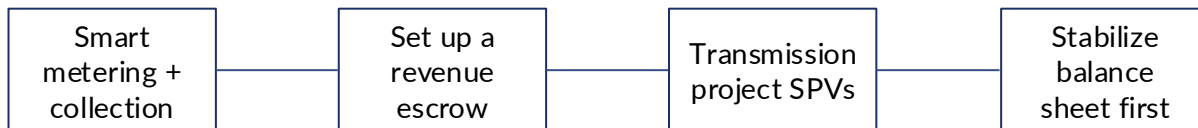
Concessional Subordinated Debt	Tranche 2	
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Senior DFI / Commercial Debt	Tranche 3	Revenue Escrow
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Optional Private Equity	Tranche 4	
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Risk ↑
Most senior ↓

BEST PRACTICE:



Thank you for listening!

Let's dive into Q&A

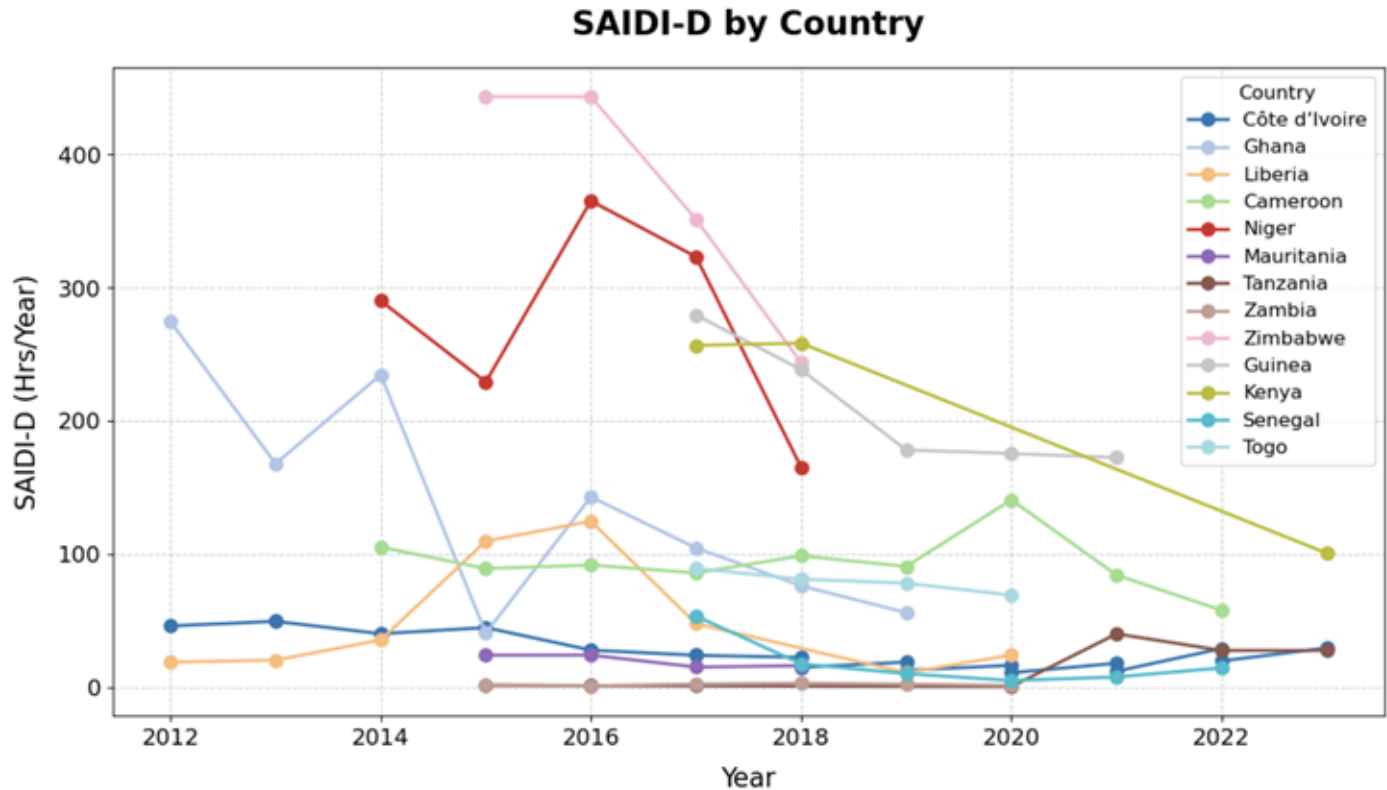
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Appendix: Tier 1 Table & Research Compacts

01		02		03		04
Generation Adequacy Indicators	+	Political Commitment to Reliability	+	Regulatory Autonomy	+	Public Demand/Opinion
→ Installed capacity (MW) → Available capacity (MW) → Peak demand (MW) → Technical losses (%) → Load-shedding (hours/month) → SAIFI/SAIDI		Assessing if Reliability is a high priority through Energy Compacts. High – Moderate – Low Specific Target → No Mention		Assessing the independence of the regulator to assess the realities on ground – can they implement change? High - Moderate - Low Fully Independent → Fully Dependent		Public Polls or demand studies – was assessed to understand the readiness of the public. High – Moderate – Low Clear Study found indicating high demand → no evidence found.
05		06		07		Moving to Tier 2
Generation Adequacy Summary Synthesis	+	Data Availability and Research Constraints	+	Country Research Compacts	■ ■	High Political Commitment + High performance in other areas [+ added benefit if high-performing state has an established relationship with WB]
The Generation Adequacy Summary column synthesizes all available quantitative indicators into a short analytical judgment.		High – Moderate – Low One or no missing indicators; data easily accessible → More than three missing.		The reliability commitment findings were added to a compact as well as any other relevant information that did not fit the scope of the table but could influence the decision making.		

Appendix: Full Time Series



Appendix: Regression

